

■ 3.2.6 Elementary Transcendental Functions

<code>Exp[z]</code>	exponential function e^z
<code>Log[z]</code>	logarithm $\log_e(z)$
<code>Log[b, z]</code>	logarithm $\log_b(z)$ to base b
<code>Sin[z], Cos[z], Tan[z], Csc[z], Sec[z], Cot[z]</code>	trigonometric functions (with arguments in radians)
<code>ArcSin[z], ArcCos[z], ArcTan[z], ArcCsc[z], ArcSec[z], ArcCot[z]</code>	inverse trigonometric functions (giving results in radians)
<code>ArcTan[x, y]</code>	the argument of $x + iy$
<code>Sinh[z], Cosh[z], Tanh[z], Csch[z], Sech[z], Coth[z]</code>	hyperbolic functions
<code>ArcSinh[z], ArcCosh[z], ArcTanh[z], ArcCsch[z], ArcSech[z], ArcCoth[z]</code>	inverse hyperbolic functions

Elementary transcendental functions.

Mathematica gives exact results for logarithms whenever it can. Here is $\log_2 1024$.

```
In[1]:= Log[2, 1024]
Out[1]= 10
```

You can find the numerical values of mathematical functions to any precision.

```
In[2]:= N[Log[2], 40]
Out[2]= 0.6931471805599453094172321214581765680755
```

$\log(-2)$ gives a complex number result.

```
In[3]:= N[ Log[-2] ]
Out[3]= 0.693147 + 3.14159 I
```

Mathematica can evaluate logarithms with complex arguments.

```
In[4]:= N[ Log[2 + 8 I] ]
Out[4]= 2.10975 + 1.32582 I
```

The arguments of trigonometric functions are always given in radians.

```
In[5]:= Sin[Pi/2]
Out[5]= 1
```

You can convert from degrees by explicitly multiplying by the constant `Degree`.

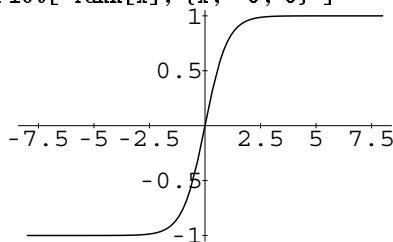
```
In[6]:= N[ Sin[30 Degree] ]
Out[6]= 0.5
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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Here is a plot of the hyperbolic tangent function. It has a characteristic “sigmoidal” form.

```
In[7]:= Plot[ Tanh[x], {x, -8, 8} ]
```



There are a number of additional trigonometric and hyperbolic functions that are sometimes used.

The **versine** function is defined as $\text{vers}(z) = 1 - \cos(z)$. The **haversine** is simply $\text{hav}(z) = \frac{1}{2}\text{vers}(z)$.

The complex exponential e^{ix} is sometimes written as $\text{cis}(x)$.

The **gudermannian function** is defined as $\text{gd}(z) = 2 \tan^{-1}(e^z) - \frac{\pi}{2}$. The **inverse gudermannian** is $\text{gd}^{-1}(z) = \log(\sec(z) + \tan(z))$. The gudermannian satisfies such relations as $\sinh(z) = \tan(\text{gd}(x))$.